ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Cumulative Frequency Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CUMULATIVE FREQUENCY DIAGRAMS · CH6 · L17

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The Map

This lesson collapses Cumulative Frequency Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Build cumulative totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Read median and quartiles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Find IQR

TRIGGER *"work out", "find", a missing value*

04 Use percentages from the curve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use cumulative frequency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 24 website questions, 24 checked solutions, 3 local PDFs read.

01 Build cumulative totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$Q_1 \text{ is at cumulative frequency } 20$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q12

There are people
This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

$$Q_1 \text{ is at cumulative frequency } 20$$

Finish: IQR about 18 minutes, and about 70 men.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 12 A total of 80 men and women took part in a race. The cumulative frequency graph gives information about the times, in minutes, they took for the race. 80 70 60 50 40 30 20 10 020 30 40 50 60 70 Time (minutes) Cumulative frequency (a) Use the graph to find an estimate for the interquartile range. minutes (2) 60% of the men took 50 minutes or less for the race. No women took 50 minutes or less for the race. (b) Work out an estimate for the number of men who took part in the race. (3)

YOUR SOLUTION

02 Read median and quartiles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{median} \approx 22.5 \text{ minutes}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q12

There are people
This is a reasoning step: write one clear sentence, then continue the calculation.

The median is at cumulative
This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: median about 22.5 minutes, IQR about 12 minutes. Hospital A is quicker and more consistent.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4$, $3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 12 The cumulative frequency graph gives information about the waiting times, in minutes, of people with appointments at Hospital A. 60 50 40 30 20 10 0 50 40 30 20 10 0 Waiting time (minutes) Cumulative frequency (a) Use the graph to find an estimate of the median waiting time at Hospital A. minutes (1) (b) Use the graph to find an estimate of the interquartile range of the waiting times at Hospital A. minutes (2) At a different hospital, Hospital B, the median waiting time is 28 minutes and the interquartile ...

YOUR SOLUTION

03 Find IQR

TRIGGER *"work out", "find", a missing value*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{median} \approx 146 \text{ g}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q11

Use cumulative frequency
This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

$$\text{median} \approx 146 \text{ g}$$

Finish: median about 146 g, and $d \approx 150$.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets. 120 90 80 70 60 50 40 30 20 10 0 Weight (grams) Cumulative frequency 130 140 150 160 170 180 (a) Find an estimate for the median of the weights of these bags of sweets. grams (2) Roberto sells the bags of sweets to raise money for charity. Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag. The total amount of money he receives by selling all the large bags is 93.75 ...

YOUR SOLUTION

04 Use percentages from the curve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

> Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** May 2025 4WM2HR Q11
>
> **Required**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Median at the th value**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** Alison is more consistent — her IQR (15 km) is smaller than Tomas's (28 km), meaning less variability in daily distances.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4$, $3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 11 The cumulative frequency graph shows information about the distance, in kilometres, Ehrick drives each day for 60 days. 10 0 20 30 40 50 60 50 40 30 60 70 80 distance (kilometres) Cumulative frequency 90 (a) Use the graph to find an estimate for the median of the distances. kilometres (1) (b) Use the graph to find an estimate for the interquartile range of the distances. kilometres (2) (c) Work out the percentage of days that Ehrick drives for more than 75 kilometres. Give your answer correct to one decimal ...

YOUR SOLUTION

05 Use cumulative frequency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$Q_1 \text{ is at cumulative frequency } 20$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q12

There are people

This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

$$Q_1 \text{ is at cumulative frequency } 20$$

Finish: IQR about 18 minutes, and about 70 men.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4$, $3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 12 A total of 80 men and women took part in a race. The cumulative frequency graph gives information about the times, in minutes, they took for the race. 80 70 60 50 40 30 20 10 020 30 40 50 60 70 Time (minutes) Cumulative frequency (a) Use the graph to find an estimate for the interquartile range. minutes (2) 60% of the men took 50 minutes or less for the race. No women took 50 minutes or less for the race. (b) Work out an estimate for the number of men who took part in the race. (3)

YOUR SOLUTION

06 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7,\ 33,\ 57,\ 71,\ 78,\ 80$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q12

Complete the cumulative frequencies

$$7,\ 33,\ 57,\ 71,\ 78,\ 80$$

Plot the points

$$(0,0),(10,7),(20,33),(30,57),(40,71),(50,78),(60,80)$$

Finish: cumulative frequencies $7, 33, 57, 71, 78, 80$, median about 23 minutes, probability about 0.095.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4$, $3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 12 The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket. Time taken (t minutes) Frequency 0 < t < 10 7 10 < t < 20 26 20 < t < 30 24 30 < t < 40 14 40 < t < 50 7 50 < t < 60 2 (a) Complete the cumulative frequency table. Time taken (t minutes) Cumulative frequency 0 < t < 10 0 < t < 20 0 < t < 30 0 < t < 40 0 < t < 50 0 < t < 60 (1) (b) On the grid opposite, draw a cumulative frequency graph for your table. 0 10 20 30 40 50 60 Cumulative frequency ...

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Build cumulative totals
"work out", "show", "hence", a familiar exam phrase	Read median and quartiles
"work out", "find", a missing value	Find IQR
"work out", "show", "hence", a familiar exam phrase	Use percentages from the curve
"work out", "show", "hence", a familiar exam phrase	Use cumulative frequency
"work out", "show", "hence", a familiar exam phrase	Convert standard form

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.